1.List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

IPv4	**IPv6**
1. Uses 32-bit addresses.	1. Uses 128-bit addresses.
2. Example: 192.168.1.10	2. Example: 2001:db8::1
3. Supports about 4.3 billion addresses.	3. Supports a much larger number of addresses.
4. Written in decimal numbers separated by dots (.).	4. Written in hexadecimal numbers separated by colons (:).
5. NAT is commonly used to share public IPv4 addresses.	5. NAT is generally not required because of its huge address space.
6. Supports broadcast communication.	6. Does not support broadcast; uses multicast instead.
7. Routers can fragment packets when needed.	7. Only the sending device can fragment packets.

2.Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

- **Network: 192.168.10.0/24**

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3.Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Step 1: Subnet mask

- **CIDR: /27**
- **Subnet mask: 255.255.255.224**
- **Borrowed bits: (27-24=3\)**
- **Host bits remaining: (32-27=5\)**

Step 2: Calculate subnets and hosts

- **Number of subnets: (2^3 = 8\)**
- **Total IP addresses per subnet: (2^5 = 32\)**
- **Usable hosts per subnet: (32-2 = 30\)**

Block size = (256-224=32)

Subnet	Network address	Usable host IPs	Broadcast address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

4.You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Given:

- **IP address: 10.0.5.17**
- **Subnet mask: 255.255.255.0**

Step 1: Convert subnet mask to CIDR

255.255.255.0 = /24

Step 2: Calculate network address

Network address: 10.0.5.0

Step 3: Calculate broadcast address

Set all host bits to 1.

Broadcast address: 10.0.5.255

Step 4: Find valid host range

Exclude the network and broadcast addresses.

- First valid host: 10.0.5.1

- Last valid host: 10.0.5.254

Item	**Result**
IP address	10.0.5.17
CIDR notation	/24
Network address	10.0.5.0
Broadcast address	10.0.5.255
Valid host range	10.0.5.1 – 10.0.5.254
Usable hosts	254

5.Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

IPv4 Subnet Calculator

Result

save

IP Address:	10.0.5.17
Network Address:	10.0.5.0
Usable Host IP Range:	10.0.5.1 - 10.0.5.254
Broadcast Address:	10.0.5.255
Total Number of Hosts:	256
Number of Usable Hosts:	254
Subnet Mask:	255.255.255.0
Wildcard Mask:	0.0.0.255
Binary Subnet Mask:	11111111.11111111.11111111.00000000
IP Class:	C
CIDR Notation:	/24
IP Type:	Private
Short:	10.0.5.17 /24
Binary ID:	00001010000000000000010100010001
Integer ID:	167773457
Hex ID:	0xa000511
in-addr.arpa:	17.5.0.10.in-addr.arpa
IPv4 Mapped Address:	::ffff:0a00.0511
6to4 Prefix:	2002:0a00.0511::/48

My manual calculation matched the online subnet calculator, and both gave the same network address, broadcast address, and valid host range.